

- 5 (a) (i) Since electromagnetic radiation is emitted when an electron falls from one discrete (specific) level to another discrete level / discrete energy levels, [1]

the photons emitted have discrete (specific) amount of energy, hence specific wavelengths since $E = hf$. [1]

(ii) $hc/\lambda = -2.5 - (-69.5) = 67.0 \text{ keV}$ [M1]

$$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (67.0 \times 10^3 \times 1.6 \times 10^{-19})$$
$$= 1.9 \times 10^{-11} \text{ m} \quad [A1]$$

- (b) 100kV spectrum shows a lower minimum wavelength (x-intercept value) compared to that of 80 kV. From $eV = hc/\lambda_{\min}$, Since V is larger, $\lambda_{\min}$ will be smaller. [1]

100 kV spectrum has higher intensity throughout since at higher voltage, electrons are emitted with higher energy. Hence, it is possible to have more collisions per unit time to produce more photons per unit time will be emitted from the tungsten. Thus increasing the intensity of the X-ray as Intensity $\propto nhf$. [1]

Characteristic lines of both spectra occur at the same wavelength since characteristic lines are formed through the release of photons when electrons move from higher energy level to lower ones. As the energy levels of Tungsten are fixed regardless of the voltage, the emitted photons have the same wavelengths. (accept same metal as explanation) [1]

- (c) Electrons behave as particles and as waves. [1]
Electron diffraction, showing wave behaviour. [1]
Electrons are deflected in electric or magnetic fields showing particle behaviour
Or electron transition between energy levels
Or Photoelectric
Or Production of X-rays
Or Collisions with atoms [1]